

Certificate E: Flow Termination

the exact time $t^* = \text{lcm}$ and the terminal cycle

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The «Dynamic Principle» Program — the hodge-laboratory

Standalone theorem monograph · Монография 13/16

1 Setting

Certificate E proves the termination of the discrete pipeline flow and determines the exact termination time. Every laboratory experiment provably stops — a fundamental stability property.

Theorem 1 (E1–E4: termination). (E1) *The state space of the flow is finite.*

(E2) *A discovery monovariant exists: every iteration either opens a new state or moves the flow to the terminal regime.*

(E3) *By the pigeonhole principle the flow terminates; the terminal state is a cycle.*

(E4) *The exact termination time is $t^* = \text{lcm}(W / \gcd(a, W), H / \gcd(b, H))$.*

Proof. (E1) States are (position, phase) pairs on a finite lattice with a finite phase group: at most $4WH$ states. (E2) The monovariant is the set of visited cells: strictly growing in the discovery regime, bounded by the grid size. (E3) Strict growth of a finite monovariant stops; the stop is a cyclic repetition of the configuration by the definition of the transition. (E4) The horizontal return is after $W / \gcd(a, W)$ steps, the vertical after $H / \gcd(b, H)$; the joint return is the LCM. Substituting the reference $(48 \times 48, \text{step } (1, 1))$ gives $t^* = 48$ — and the flow terminates in exactly 48 steps with the full coordinate cross-check $48/212/432/114$. $\square$

Protocol data: the eight protocol points

E-1: scanning 64 step pairs — the monovariant grows, the flow terminates. E-2: the 48×48 pipeline: $t^* = 48$; the $48/212/432/114$ cross-check complete. E-3: the $(7, 22)$ correction step — termination under the correction rules. E-4: the terminal cycle of length 4 — the μ_4 -orbit of the step. E-5: scales $48 \rightarrow 384$ — the t^* formula persists. E-6: the K3 layer (reflections with braking) terminates. E-7: the Klein layer: Singer generation of order 7. E-8: the Klein layer: μ_4 -orbits of length 4. Verdict: 8/8 PASS.

Remark 2 (a cycle — a theorem, not a hypothesis). The cyclic structure of the terminal state follows from finiteness and step periodicity; the cycle length 4 is exactly the μ_4 -orbit. This is a theorem about the *structure* of the terminal regime, not merely its existence.

Summary of results: summary

Certificate E accepted (8/8): the flow terminates, the termination time is computed exactly, the terminal structure is described.

Verification: reproduction

`python3 laboratory.py` → “Certificates → E”; the designer: an arbitrary $W \times H$ grid and step (a, b) — the lab prints t^* and verifies by enumeration.

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